

How to Include Styles?

- **Inline Styles**

Writing style directly inline on each element

`<h1 style="color: red"> Apna College </h1>`

→

```
<> index.html > html > body
1  <!DOCTYPE html>
2  <html lang="en">
3    <head>
4      <meta charset="UTF-8" />
5      <meta http-equiv="X-UA-Compatible" content="IE=edge" />
6      <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7      <title>CSS Chapter 1</title>
8    </head>
9    <body>
10     <!-- Inline Style -->
11     <h1 style="color: red">Apna College</h1>
12     <p style="color: pink">This is a paragraph.</p>
13     <button style="color: blue; background-color: orange">Help</button>
14   </body>
15 </html>
16
```

How to Include Styles?

- Using <style> tag

Style is added using the <style> element in the same document

reddy2096@gmail.com

```
<style>
  h1 {
    color : red;
  }
</style>
```

```
<> index.html > html > body
1  <!DOCTYPE html>
2  <html lang="en">
3    <head>
4      <meta charset="UTF-8" />
5      <meta http-equiv="X-UA-Compatible" content="IE=edge" />
6      <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7      <title>CSS Chapter 1</title>
8      <style>
9        h1 {
10          color: red;
11        }
12      </style>
13    </head>
14    <body>
15      <!-- Inline Style -->
16      <!-- <h1 style="color: red">Apna College</h1>
17      <h1 style="color: red">Shradha Khapra</h1>
18      <h1 style="color: red">Hello World!</h1>
19      <p style="color: pink">This is a paragraph.</p>
20      <button style="color: blue; background-color: orange">Help</bu
21
22      <!-- Style Tag -->
23      <h1>Apna College</h1>
24      <h1>Shradha Khapra</h1>
25      <h1>Hello World!</h1>
26    </body>
27  </html>
```

How to Include Styles?

- **External Stylesheet**

Writing CSS in a separate document & linking it with HTML file

↓
css file

Linking HTML with CSS File

<head>

<link rel="stylesheet" href="style.css">

</head>

```
index.html x # style.css jcreddy209
index.html > html > body
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8" />
5   <meta http-equiv="X-UA-Compatible" content="IE=edge" />
6   <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7   <title>CSS Chapter 1</title>
8   <link rel="stylesheet" href="style.css" />
9 </head>
10 <body>
11   <!-- Inline Style -->
12   <!-- <h1 style="color: red">Apna College</h1>
13   <h1 style="color: red">Shradha Khapra</h1>
14   <h1 style="color: red">Hello World!</h1>
15   <p style="color: pink">This is a paragraph.</p>
16   <button style="color: blue; background-color: orange">Help</button> -->
17
18   <!-- Style Tag -->
19   <h1>Apna College</h1>
20   <h1>Shradha Khapra</h1>
21   <h1>Hello World!</h1>
22 </body>
23 </html>
24
```

Rel: tells the relation css is a cascading style sheet know

Href: here we put the name if it is in the same folder if not in same folder then we need to put the path.

Color Property

Used to set the color of foreground

color : purple;

color : #ffffff;

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8" />
5     <meta http-equiv="X-UA-Compatible" content="IE=edge" />
6     <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7     <title>CSS Chapter 1</title>
8     <link rel="stylesheet" href="style.css" />
9   </head>
10  <body>
11    <h1>Apna College</h1>
12    <h1>Shradha Khapra</h1>
13    <p>This is paragraph 1</p>
14    <p>This is paragraph 2</p>
15    <button>Button 1</button>
16    <button>Button 2</button>
17  </body>
18 </html>
```

```
style.css > h1
1 h1 {
2   color: blue;
3 }
4
5 p {
6   color: red;
7 }
8
9 button {
10  color: gold;
11 }
12
```

Apna College

Shradha Khapra

This is paragraph 1

This is paragraph 2 @gmail.com

Button 1

Button 2

Background Color Property

Used to set the color of background

```
background-color : purple;
```

```
background-color : #ffffff;
```

```
# style.css > button
1  h1 {
2    color: blue;
3    background-color: gray;
4  }
5
6  p {
7    color: red;
8    background-color: cornsilk;
9  }
10
11 button {
12   color: gold;
13   background-color: black;
14 }
15
```

creddy2096@gmail.com

Color Systems

RGB

→ 1st

Red Green Blue (0-255)

red is `rgb(255, 0, 0)`

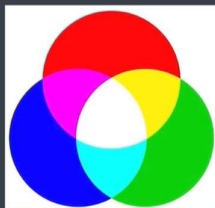
black is `rgb(0, 0, 0)`

blue is `rgb(0, 0, 255)`

yellow is `rgb(255, 255, 0)`

`color: rgb(255, 0, 0)`

`color = rgb(0, 0, 0);`



Red channel
Green channel
Blue channel



```
style.css > h2
h2 {
  color: rgb(0, 0, 0);
}
```

Text properties

Text Properties

text-align

font-weight

text-decoration

line-height

letter-spacing

font-size

1. text align

text-align



text-align: left / start

text-align: right / end

text-align: center

text-align: justify

2. font weight

font-weight *→ light* *→ bold* 100 — 900

font-weight: normal //400

font-weight: bold //700

font-weight: 100 ✓

{ font-weight: bolder } parent
{ font-weight: lighter }



3. text decoration

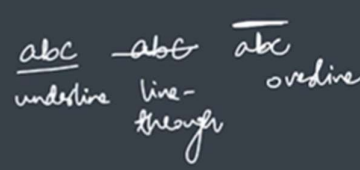
text-decoration

sets the **appearance of decorative lines** on text like underline

text-decoration: underline

text-decoration: overline

text-decoration: line-through



text-decoration

text-decoration: red underline

text-decoration: wavy overline

text-decoration: dotted line-through

✓ CSS Property: `text-transform`

CSS

```
text-transform: uppercase;
```

4. line height

line-height

controls the **height** of the line of text

line-height: normal ✓

line-height: 2.5 ✓

2.5 =

5. letter spacing

letter-spacing

controls the **horizontal spacing** behavior between text characters

letter-spacing: normal

letter-spacing: 10px
=

abc

a ↔ b ↔ c

px = pixels

→font-size units in css

5cm

Font-size units in CSS	
Absolute	Relative
px	%
pt	em
pc	rem
cm	ch
mm ✓	vh
in	vw + many more

1 inch = 96pixels

Pixels (px)

most commonly used absolute unit

96px = 1 inch

not suitable for responsive websites

```
font-size: 50px ;
```

font-family → type

specifies a prioritized list of one or more font family names

```
font-family: arial ;
```

```
font-family: avant garde, didot, sans-serif ;
```

backup family
of fonts



→ selectors:

Universal Selector

To select everything in a document

Universal
Selector

```
* {
```

```
    property: value;
```

```
}
```

universal selector is used to apply the changes to all the elements in the page with the help of *

→ element selector:

Element Selector

Selects all elements of the same type

Element
Selector

`h1 {`
`property: value;`
`}`

element selector:

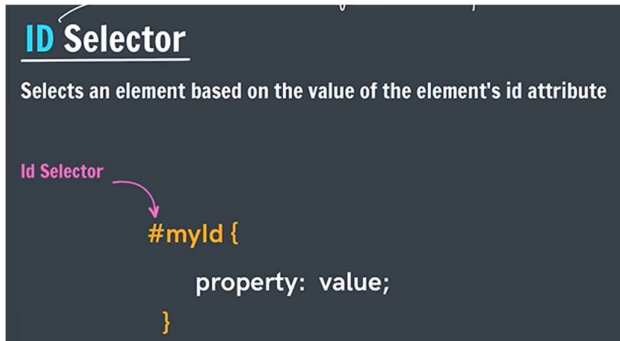
we select the individual different tags and apply the styling to them.

```
style.css > *  
1 * {  
2   font-family: "Courier New";  
3 }  
4  
5 h1, h3 {  
6   color: #b92b27;  
7 }  
8
```

```
style.css > h1  
1 * {  
2   font-family: "Courier New";  
3 }  
4  
5 h3 {  
6   color: purple;  
7 }  
8  
9 h1 {  
10  color: #b92b27;  
11 }  
12
```

we can write multiple selectors with separating with commas and can give the same styling to them.

→id selector

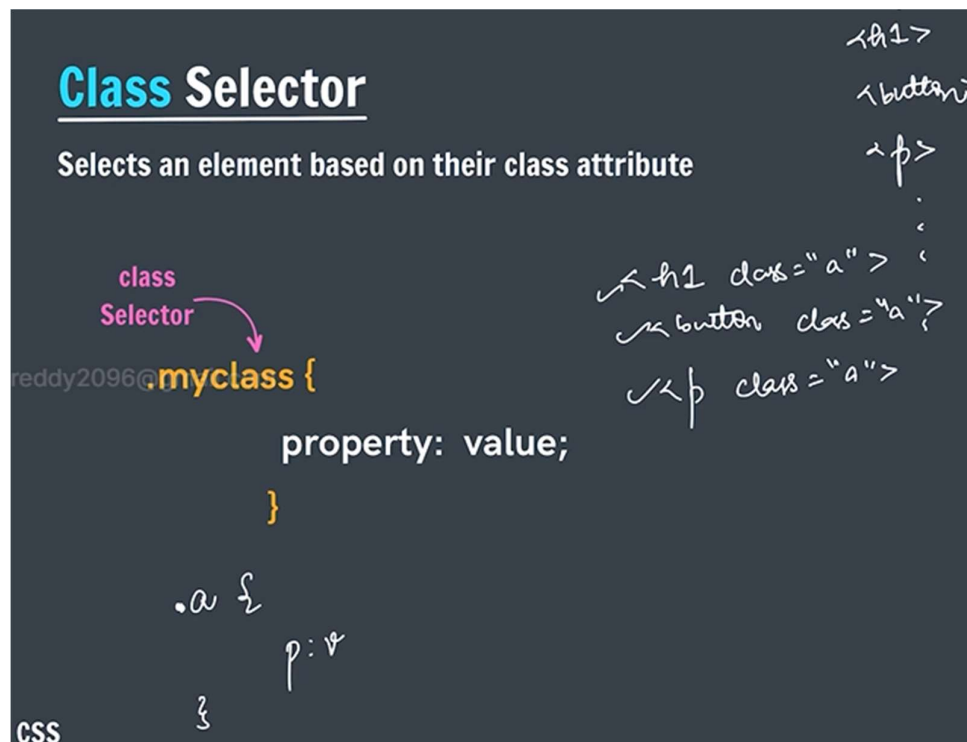


id should be given unique for the each element and we give the styling by selecting that id.

Below is the example of id selectors.

```
#login  
{  
background-color: white;  
}  
#signup  
{  
background-color: #b92b27;  
color: white;  
}
```

→class selectors:



```
{
  background-color: green;
}
```

here we use the different attributes

→ Descendent selector

Eg - Selects all paragraphs inside divs

```
div p {  
    property: value;  
}
```

descendent selector means it will give the styling to the elements which is inside ones

ex:if there are multiple paragraph tags in that inside if we have anchor tag to apply style to the anchor tag we use this

```
/* descendant selector */
p a {
  color: purple;
  background-color: yellow;
}
```

->if we group the required paragraph tag with the same class name then we use it like the .class a inside this we can apply the styling.

```
/* descendant selector */
.post a {
  color: purple;
  background-color: yellow;
}
```

- [Home](#)
- [Following](#)
- [Answer](#)
- [Spaces](#)

```
nav ul li {
  color: hotpink;
}
```

username

- [Home](#)
- [Following](#)
- [Answer](#)
- [Spaces](#)

Selector Specificity: (0, 0, 0, 0)

```
nav ul li a {
  color: hotpink;
}
```

→adjacent sibling combinator

APNA
COLLEGE

APNA COLLEGE

Paragraph →

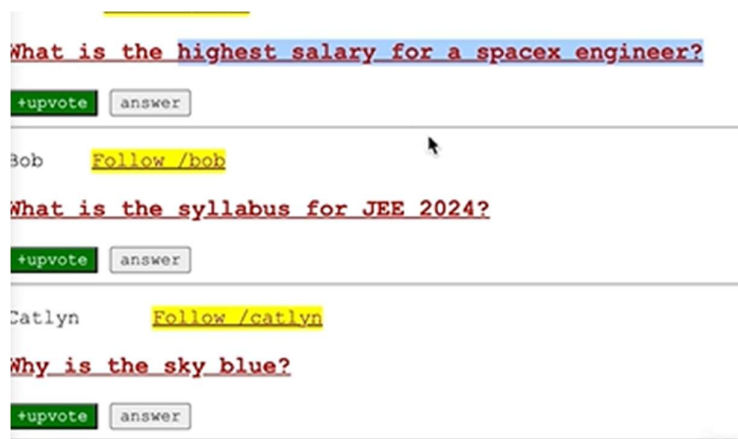
- $\langle p \rangle \rightarrow \langle /p \rangle$
- $\langle h_3 \rangle \rightarrow \langle /h_3 \rangle$
- $\langle h_3 \rangle \rightarrow \langle /h_3 \rangle$
- $\langle d.o \rangle \rightarrow \langle /d.o \rangle$

$p \rightarrow h_3$

$h_3 \times$, $p \times$

$d.o \downarrow h_3$

```
p + h3 {  
  ↑      ↑  
  l      l  
  property: value;  
}
```

[illegible]

→ child combinator:

Child Combinator

Eg - Selects all buttons which are direct children of spans

eddy2child@gmail.com

Selector

```
span > button {  
    property: value;  
}
```



`<div>` `<p>` `</p>` `</div>`
div
↓
p child
↓
ul
↓
li
direct descendant

Difference between Descendant Selector and Child Combinator (CSS)

1. Descendant Selector

Definition:

A descendant selector selects **all matching elements that are inside a parent element**, at **any level** (child, grandchild, etc.).

Syntax:

parent child

Example:

```
div p {  
    color: red;  
}
```

Explanation:

All `<p>` elements inside `<div>` will be selected, even if they are deeply nested.

2. Child Combinator

Definition:

A child combinator selects **only the direct (immediate) children** of a parent element.

Syntax:

parent > child

Example:

```
div > p {
```



```
color: blue;  
}
```

Explanation:

Only <p> elements that are **direct children** of <div> will be selected.

Key Differences (Write This Table)

Feature	Descendant Selector	Child Combinator
Symbol	Space ()	Greater than (>)
Selection level	Any level inside parent	Only direct children
Grandchildren selected	Yes	No
Selection type	Broad / loose	Strict / specific
Use case	When structure depth is unknown	When exact structure is known

One-Line Exam Answer

Descendant selector selects all matching elements inside a parent at any level, whereas child combinator selects only direct children of a parent element.

→ attribute selector:

Attribute Selector

Selects elements based on the presence or value of a given attribute

Attribute
Selector

```
input[attr="value"] {  
    property: value;  
}
```

```
input[type="text"] {  
    <input type="password">  
    Selector Specificity: (0, 1, 1)  
    input[type="password"] {  
        color: red;  
    }  
  
    input[type] {  
        background-color: ghostwhite;  
    }  
}
```

->Pseudo class

Pseudo Class

element : pseudoclass

A Keyword added to a selector that specifies a **special state** of the selected element(s)

: hover

: active

: checked

: nth-of-type

o yes

o no

jcreddy209

```
button:hover {  
    background-color: black;  
    color: white;  
}
```

What is the high

What is the high

+upvote

answer

+upvote

answer

here

hover means if we take the cursor to the button it colour changes

→ active: this will work when we click the button then it changes its properties.



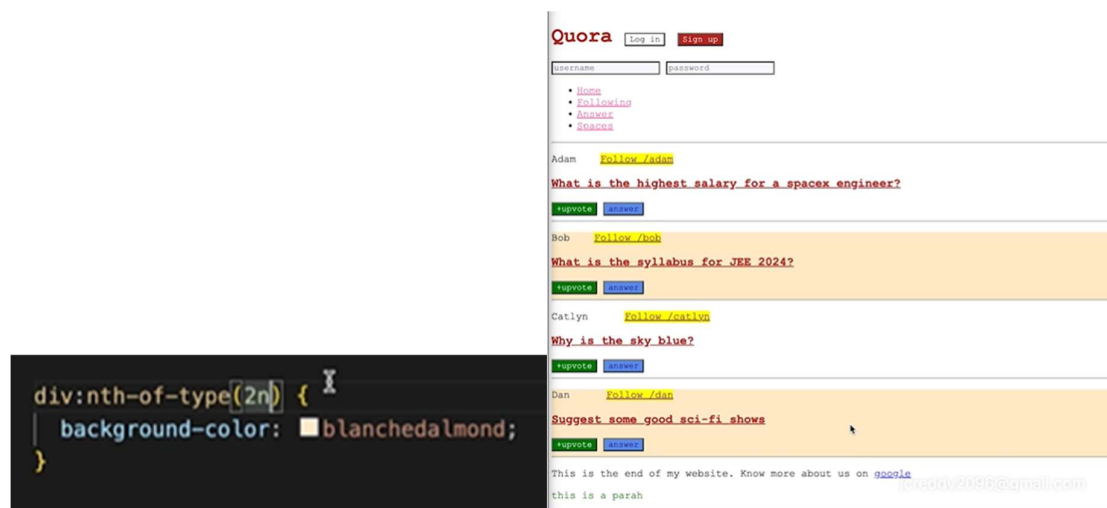
What is the high



→ checked: in this we can use like if we click on no then it will change its properties as given



Nth of type:



Here by the use nth of type we can change the styling properties by using this.

Pseudo elements:

Pseudo Elements

A Keyword added to a selector that lets you style a **specific part** of the selected element(s)

:: first-letter

:: first-line

:: selection



->first-line:



This is the end of my website. Know more about us on [google](#)
Lorem ipsum dolor sit amet consectetur adipisicing elit. Distinctio iure totam, ipsum sit alias maxime, corrupta laborum nobis, dolor incididunt rem! Dolorum explicabo modi sint ipsa.

Selection:



→css:cascading stylesheets

CSS : Cascading StyleSheets

What is cascade in CSS?

The **CSS cascade algorithm**'s job is to select CSS declarations in order to determine the correct values for CSS properties.

```
h2 {  
    background-color: yellow;  
}  
  
h2 {  
    background-color: blue;  
}
```

final color is blue

→this property tells that the which is written last or executed at last that value will be consider here above h2 have first one have yellow and 2nd one have blue the color will be blue.

→if I used the two css files in that which file will be executed last that file property will be applied

```
style2.css > h3  
h3 {  
    background-color: green;  
}
```

```
<title>CSS</title>  
<link rel="stylesheet" href="style.css" />  
<link rel="stylesheet" href="style2.css" />  
</head>
```



→ selector specificity:

Selector Specificity

What is specificity?

Specificity is an algorithm that calculates the weight that is applied to a given CSS declaration.

id

class,
attribute &
pseudo-class

element &
pseudo-
element

If specificity is same then we see cascading.

Practice Qs

For the given html & css code, what is color of text "Hello World!":

```
<div class="greetDiv">
  <p class="greet">
    Hello World!
  </p>
</div>
```

```
div.greetDiv .greet {
  color: green;
}

.greetDiv p.greet {
  color: blue;
}

.greetDiv .greet {
  color: red;
}
```

specificity → cascading APNA COLLEGE

blue
↑

$\frac{0}{div} \frac{2}{class} \frac{1}{element} (21)$
 $\frac{0}{div} \frac{2}{class} \frac{1}{element} (21)$
 $\frac{0}{div} \frac{2}{class} \frac{0}{element} (20)$



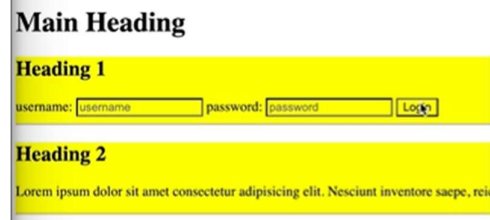
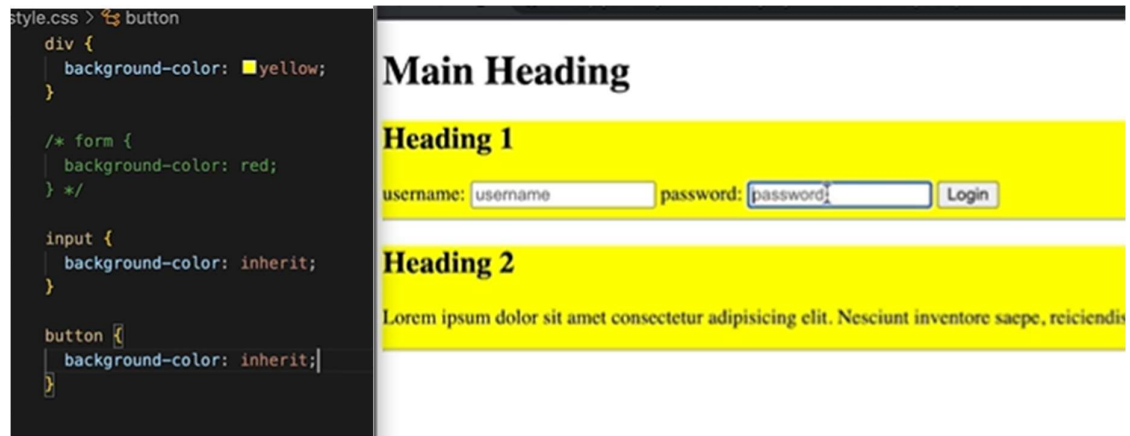
→important

!important

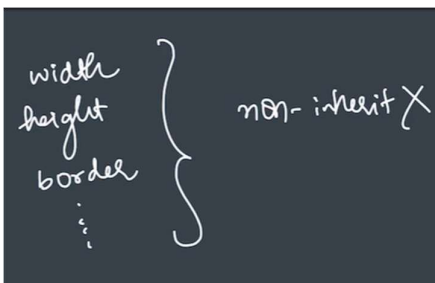
To show the most specific thing in document

```
h2 {  
    background-color: blue !important;  
}
```

→inheritance in css:



here some input fields and buttons doesn't inherit the colours automatically so we use this to get colour



Width,height,border veetiki inherit kavu.